EE696 Interim Report

Project Title: GrowMate: Transparent Voice-Robot Interaction through LLM-Constructed Behaviour Trees for Accessible Agricultural Robotics

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Abstract

Agricultural robots such as FarmBot can automate gardening tasks, but their control inter- faces remain inaccessible to non-expert users, particularly elderly and mobility-impaired indi- viduals. This report presents GrowMate, a voice-controlled assistant for FarmBot that uses an edge large language model (LLM) to classify user intent from natural speech and then constructs inspectable behaviour trees (BTs) from a typed node library. Where existing approaches map speech to robot commands through opaque inference, GrowMate produces transparent execution plans made up of robot action nodes, external function calls, LLM reasoning steps, and safety condition nodes. A dedicated BT engine executes these plans node by node, validating safety at each step through workspace bounds checking, plant existence verification via the FarmBot REST API, and user confirmation dialogs for high-impact operations. Preliminary evaluation on 29 test utterances using Gemma 3 4B via Ollama shows a Desired Behavior Success Rate of 96.6%, a Single Node Success Rate of 98.8%, zero unsafe state occurrences, and 100% tree structural validity. The system integrates with the AURA FarmBot ROS2 stack as a drop-in voice controller and requires no modifications to the existing robot software.

Declaration

I hereby declare that this report is entirely my own work, except where otherwise stated. I have not plagiarised the work of others or colluded with others. All sources have been properly cited and referenced.

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Use of Generative AI: AI tools (Claude, Anthropic) were used for code development assis- tance, prompt engineering iteration, and document structuring. All technical decisions, system architecture, and evaluation methodology are the author's own work. This use is disclosed per Maynooth University GenAI guidelines and IEEE policy.

Contents

1 Introduction **3**
1.1 Motivation . . . 3
1.2 Problem Statement . . . 3
1.3 Approach Overview . . . 4
1.4 Ethical Considerations and Broader Impact . . . 4

2 Technical and Related Background **4**
2.1 Voice Control for ROS-Based Robots . . . 4
2.2 Behaviour Trees in Robotics . . . 5
2.2.1 Evaluating Behaviour Trees . . . 5
2.3 LLM-Based Behaviour Tree Generation . . . 6
2.4 LLM Safety in Robotic Systems . . . 6
2.5 On-Device LLMs for Edge Robotics . . . 6
2.6 Agricultural Robot Interfaces and Accessibility . . . 7
2.7 Summary and Research Pathway . . . 7

3 Methodology **7**
3.1 System Architecture . . . 7
3.2 Hybrid Intent Classification and Tree Construction . . . 8
3.3 Node Library and BT Engine . . . 8
3.4 Evaluation Setup . . . 9

4 Progress and Work Plan **9**
4.1 Work Package Overview . . . 9
4.2 Gantt Chart . . . 10
4.3 Completed Work . . . 10
4.4 Preliminary Results . . . 11

1 Introduction

1.1 Motivation

FarmBot is an open-source precision agriculture system that automates planting, watering, and soil monitoring for raised-bed gardens [1]. At Maynooth University, the AURA team has built a ROS2 control stack for it, adding modular control and multi-robot coordination [2]. The platform, at present, can only be operated through terminal commands over SSH. That is a high bar for most people. It is an impossible bar for the people who might benefit most.

A focus group conducted jointly between Maynooth University and Dundalk Institute of Technology brought together 18 older adults (mean age 72.4 years) to discuss gardening challenges and robotic assistance [3]. Mobility was the central obstacle. Eight participants reported health issues, including arthritis and back pain, that limit or prevent their gardening. Several described a prolonged loss of quality of life. As one participant put it: "I loved farming... I used to love it... years, years, years." Another said: "through mobility, I had to stop doing it all... it made a big difference to me." For some, family members now look after the garden, but the standard of upkeep tends to fall short.

Technology confidence among the group was low: only 4 of 18 (22%) felt fully confident with new technology, while 13 had little or no confidence. When asked whether they would prefer voice or buttons, the preference for voice was clear. One woman explained, half-laughing, that her arthritis made buttons "not an option, love." Several drew comparisons to Amazon Alexa without being prompted, with one asking: "would it be an idea if you could speak to the equivalent of Alexa and say look for advice?" What surprised us was how many participants wanted the system to *know things*, not just do things. They asked about planting schedules, pest identification, soil conditions. Watering came up most often, tied to the weight of watering cans.

1.2 Problem Statement

When a large language model translates natural language into robot commands, the reasoning behind those commands is not visible. If the system produces `[M 400 200 -100, D_W_1, D_W_0]` in response to "water the tomatoes," there is no way for a user or developer to know why those particular commands were chosen, what order they should run in, or what should happen if one of them fails. For vulnerable users controlling a physical robot, this lack of visibility creates problems of both safety and trust.

Existing voice-robot control systems share this limitation. ROSGPT uses GPT-4 to translate commands into ROS2 actions but depends on cloud connectivity and treats the entire mapping as a single opaque step. ROSA, built at NASA JPL, constrains commands to predefined patterns, which limits natural conversational input. RAI provides a multi-modal ROS2 framework with cloud LLMs. Nwankwo and Rueckert show that everyday conversation can control real-world robots without a structured command vocabulary [4–7]. In each case, the system produces robot commands through inference that cannot be inspected, validated at intermediate steps, or composed with non-robot operations such as data retrieval.

We focus on the design, implementation, and evaluation of a voice-controlled interface for

the AURA FarmBot ROS2 stack. The scope covers the full voice-to-action pipeline: speech recognition, intent classification, behaviour tree construction and execution, and text-to-speech feedback. The FarmBot ROS2 software is not modified; instead, the system publishes to `keyboard_topic`, the same ROS2 topic that the existing keyboard controller uses.

1.3 Approach Overview

Rather than having the LLM generate command strings, we represent all voice-robot interactions as behaviour trees constructed from a typed node library. The LLM classifies user intent from natural speech, and deterministic code assembles the tree. A BT engine then executes the plan node by node with safety validation at each step. The full system design, including the architecture, node library, and hybrid classification approach, is described in Section 3 (Methodology). The evaluation follows the standardised BT metrics of Gugliermo et al. [10], detailed in Section 4.

1.4 Ethical Considerations and Broader Impact

Our work contributes to UN Sustainable Development Goal 2 (Zero Hunger) by supporting independent food production for people who cannot use conventional digital interfaces, and to SDG 10 (Reduced Inequalities) by making assistive agricultural technology accessible to elderly and disabled users. The focus group data shows clearly that the inability to garden has a lasting negative impact on quality of life for the older adults we spoke with [3].

All voice data is processed on-device using a local LLM, which avoids cloud dependency and the privacy concerns that come with it. Physical safety is managed through workspace bounds checking at every movement node, an emergency stop that bypasses the AI pipeline entirely, and confirmation dialogs that require user approval before high-impact operations execute. Hundt et al. [12] have documented that LLM-driven robots can produce unsafe or harmful actions, and they argue for explicit safety constraints in any LLM-robot pipeline. The per-node safety architecture addresses this concern.

A user study with elderly participants is planned for the final phase. Ethics approval will be sought from Maynooth University before any participant involvement.

2 Technical and Related Background

2.1 Voice Control for ROS-Based Robots

Natural language control of ROS-based robots has been explored through several systems in recent years, each taking a different position on the trade-off between flexibility and safety. ROSGPT translates spoken commands to ROS2 actions using GPT-4 [4]. It demonstrates feasibility but requires a persistent cloud connection and introduces variable latency. The mapping from speech to action happens in a single LLM call, with no intermediate structure or safety validation.

ROSA, built at NASA JPL, takes the opposite approach: it matches voice input against predefined patterns and applies safety rules [5]. The trade-off is that natural, unstructured input

from non-technical users is not well supported. RAI offers a multi-modal ROS2 framework that also relies on cloud LLMs [6]. Nwankwo and Rueckert showed something we found genuinely encouraging: users can control a real robot using everyday conversation, no command vocabulary required [7]. Their paper makes a convincing case that natural language is sufficient for robot control. It does not, however, address what happens when the conversation maps onto a physical action that could go wrong.

What these systems have in common is that the reasoning behind their output is invisible. A user cannot see why a particular command was chosen, cannot check safety at intermediate steps, and cannot combine robot actions with external operations like weather lookups or database queries within a single plan. For the elderly users in our focus group [3], who expressed a strong desire to understand and control what the robot does, this lack of transparency is a barrier to trust.

2.2 Behaviour Trees in Robotics

Behaviour trees are hierarchical structures for organising robot tasks. They were originally developed for controlling non-player characters in video games and have since been widely adopted in robotics for their modularity and fault tolerance [8]. A BT is a directed acyclic graph. Internal nodes control execution flow: a Sequence node runs its children in order and stops on the first failure; a Selector node tries alternatives until one succeeds. Leaf nodes either perform actions or check conditions. Every node returns one of three statuses: Success, Failure, or Running. Iovino et al. [9] survey BT applications across robotics, covering low-level control, task switching, and multi-agent coordination.

Colledanchise and Ögren [8] establish the formal foundations. Their argument for BTs over finite state machines rests on modularity: subtrees are independent modules with a common interface (the three return statuses), so they can be developed, tested, and reused independently. That modularity property turned out to matter more than we had expected. Because every node type shares the same interface, a robot action node, a function call node, and an LLM reasoning node can all sit in the same tree and be composed freely.

2.2.1 Evaluating Behaviour Trees

One challenge in the BT community is the lack of agreement on how to measure and compare BT policies. Gugliermo et al. [10] address this gap with a framework that defines both functional and non-functional metrics.

On the functional side, they define Desired Behavior Success Rate (DBSR), measuring whether the overall goal is achieved; Single Node Success Rate (SNSR), tracking success at the individual node level; and Unsafe State Count (USC), counting how often the system enters states that could lead to irreversible harm.

On the non-functional side, they measure Tree Dimensions (height, width, total node count), Action Granularity (the degree to which actions are decomposed into finer steps), and Graph Edit Distance (for comparing structural similarity between trees). Their definition of safety is especially relevant: a BT is safe if every action node, when executed from a safe state, transitions to another safe state, and if the tree design guarantees that no action runs unless its preconditions

are met. Our evaluation adopts DBSR, SNSR, USC, Tree Dimensions, and Action Granularity from their framework.

2.3 LLM-Based Behaviour Tree Generation

Lykov and Tsetserukou [11] demonstrated that an LLM can produce structurally valid behaviour trees from text descriptions. Their system, LLM-BRAIn, is a 7B parameter model fine-tuned on 8,500 BT examples to output trees in XML format. In a study with 15 robotics students, participants were unable to tell the LLM-generated trees from hand-written ones. The average correct identification rate was 4.53 out of 10, statistically indistinguishable from guessing ($t = -1.200$, $p > 0.05$). That result is striking: the structural quality is high enough that experts cannot spot the machine's work.

What LLM-BRAIn does not include is non-robot nodes. There is no way to call a weather API, query a plant database, or ask the LLM to reason about sensor data within the tree itself. The trees also execute without intermediate safety checks. We extend the approach in three ways: a node library that puts function calls and LLM reasoning alongside robot actions, safety validation at every node during execution, and zero-shot prompting with a structured node library rather than domain-specific fine-tuning.

2.4 LLM Safety in Robotic Systems

Hundt et al. [12] examined the risks of using LLMs to drive robot behaviour and found that LLMs can produce actions that are unsafe, discriminatory, or otherwise harmful. Their argument is that pipelines which pass LLM output to a robot without structural safeguards are inherently risky, particularly in environments shared with people.

This concern shaped a core design decision. Instead of treating the LLM's output as a trusted command sequence, the BT engine checks every node against its preconditions before execution. Movement commands must pass a bounds check. Commands targeting a specific plant must first verify that the plant exists in the FarmBot database. Operations affecting the entire garden require the user to say yes. If any check fails, the sequence stops and nothing reaches the robot. But we are not certain this is enough. The safety checks cover the known risks, but what about risks not yet anticipated?

2.5 On-Device LLMs for Edge Robotics

Running an LLM on a phone or embedded device, rather than in the cloud, has become practical with recent model architectures. Gemma 3n is designed for mobile deployment and achieves 60 to 70 tokens per second on mobile processors with as little as 2 GB of RAM [13]. Chandra [14] surveys the broader landscape, showing that capable models can now run on consumer hardware without network access.

For a garden robot that may operate in a greenhouse with no WiFi, this is a practical requirement. The current GrowMate prototype uses Gemma 3 4B running locally through Ollama [15]. Intent classification takes roughly 6 seconds on consumer desktop hardware. The

architecture is designed so that switching models requires only a configuration change; published benchmarks suggest Gemma 3n on a phone would bring latency below 2 seconds [13].

2.6 Agricultural Robot Interfaces and Accessibility

FarmBot's official interface is a web application that requires coordinate entry and technical menus [1]. The AURA ROS2 stack offers no graphical interface at all [2]. Brăileanu [16] surveys human-computer interaction in smart greenhouses and notes that voice-based interaction is largely absent from the field, despite its potential for accessibility.

Outside agriculture, there is relevant work on robot interaction with older adults. Blavette et al. [17] tested a socially assistive robot with an integrated LLM in a geriatric care institution and found that elderly residents could interact with it meaningfully. Their mixed-methods evaluation, combining quantitative usability measures with qualitative feedback, provides a useful template for our planned user study. Zhang et al. [19] explored AI-driven robots in garden settings, though without a voice interface aimed at non-expert users.

On the speech recognition side, Shekoufandeh et al. [18] showed that fine-tuning Whisper on the JASMIN-CGN corpus improved ASR accuracy for elderly Dutch speakers. Although their work targets a different language, the methodology of fine-tuning on age-specific speech data is transferable to the challenge of recognising elderly Irish English.

2.7 Summary and Research Pathway

Taken together, the literature shows that LLMs can produce valid behaviour trees from natural language [11]; that BTs offer an inspectable, formally evaluable execution model [8, 10]; that LLM-robot pipelines need structural safety mechanisms beyond what the model itself provides [12]; that on-device LLMs are now practical [13, 14]; and that agricultural robots lack accessible voice interfaces [1, 16]. The Dundalk focus group showed that voice control, gardening advice, and selective automation with user confirmation are the primary needs of the target population [3].

GrowMate brings these threads together: an edge LLM classifies intent from natural speech, deterministic code constructs inspectable behaviour trees from a typed node library, and a BT engine executes them with safety validation at every node.

3 Methodology

3.1 System Architecture

GrowMate follows a three-layer architecture, illustrated in Figure 1. The User Interaction Layer captures speech through a microphone and delivers spoken feedback. The AI Processing Core classifies intent and constructs behaviour trees. The Robot Control Layer publishes validated commands to `keyboard_topic`, the same ROS2 topic that FarmBot's own keyboard controller uses.

Emergency commands ("stop," "halt," "freeze") skip the AI layer entirely and go straight to the robot controller. We measured the bypass latency at under 1 ms.

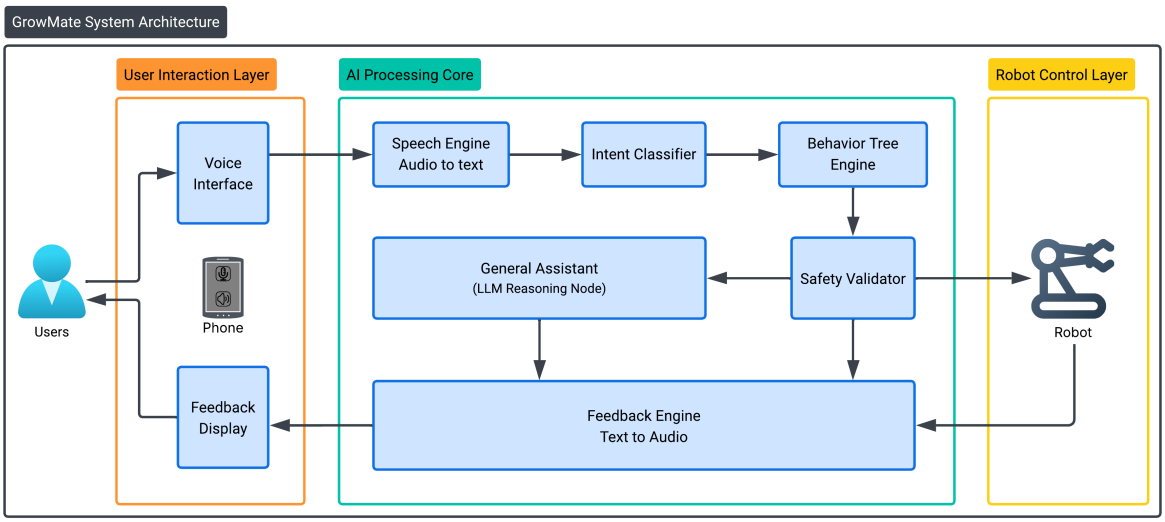

Figure 1: GrowMate system context diagram. Speech enters at the left. The AI Processing Core classifies intent and constructs behaviour trees. Commands exit through the Robot Control Layer to FarmBot via ROS2.

3.2 Hybrid Intent Classification and Tree Construction

Separating intent classification from tree construction came from a practical finding during development. Small edge LLMs in the 2–4B parameter range classify intent reliably but produce malformed output when asked to generate nested JSON tree structures. We found this out early in development: initial experiments with direct tree generation from Gemma 3n produced valid JSON roughly zero percent of the time. By giving the LLM the simpler task and letting code handle the structural complexity, the system gets both natural language understanding and structural correctness.

The LLM receives the transcribed speech and returns a structured classification: the action type, the target plant or location, and a spoken response for the user. This classifier supports multiple intents per utterance, so a request like "water the tomatoes and then go home" produces two classified actions in sequence. Based on that classification, deterministic Python code constructs a behaviour tree from the typed node library. The resulting tree contains robot action nodes that generate FarmBot commands, function call nodes that query the FarmBot API or weather services, LLM reasoning nodes that analyse gathered data, safety condition nodes that validate preconditions, and response nodes that produce spoken feedback.

3.3 Node Library and BT Engine

The BT engine operates on a typed node library of 29 node types across six categories. Each node receives parameters, executes a single operation, and returns Success, Failure, or Running. Robot action nodes (12 types) map to FarmBot command strings. Function call nodes (6 types) query external services such as the FarmBot REST API and weather APIs, storing results as named variables. LLM reasoning nodes (2 types) pass accumulated context to the language model and store the response. Safety condition nodes (4 types) check preconditions: `check_available` verifies system readiness, `check_bounds` validates workspace coordinates, `check_plant_found`

confirms plant existence, and `confirm` pauses for user approval on broad-scope operations. Utility nodes (3 types) handle waiting, variable assignment, and logging. Control nodes (sequence, selector) govern execution flow.

Results from function call nodes propagate to subsequent nodes through dot-notation variable resolution (e.g., `$plant_data.x`), allowing the tree to bind live data from the FarmBot API at runtime. If the API is unreachable, the system falls back to a YAML configuration file.

3.4 Evaluation Setup

The system is evaluated against a corpus of 29 natural language inputs across seven categories: direct robot commands (10), indirect or implied commands (5), sensor queries (4), general knowledge questions (3), multi-step tasks (2), safety-critical commands (1), and emergency stops (4). All tests use simulated robot execution against a garden configuration with five plant types and three named locations. The evaluation uses Gemma 3 4B running locally through Ollama on consumer desktop hardware.

Both functional properties (correctness, safety, latency) and non-functional properties (tree validity, dimensions, action granularity) are measured using the framework of Gugliermo et al. [10]. DBSR measures whether the tree achieves the user's goal. SNSR tracks individual node success. USC counts entries into unsafe states. Tree Dimensions and Action Granularity characterise structural properties.

4 Progress and Work Plan

4.1 Work Package Overview

We organised the work into five work packages.

WP1: Research and System Design (Jan to Mar 2026). Literature review, architecture design, FarmBot ROS2 command analysis, and node library specification. **Complete.**

WP2: Core Implementation (Feb to Apr 2026). AI Core with multi-intent classifier, BT Engine with per-node safety and FarmBot API integration, Robot Controller for ROS2 publishing. **Complete.**

WP3: Evaluation and Refinement (Mar to May 2026). Evaluation framework, test corpus of 29 utterances across 7 categories, baseline evaluation on Gemma 3 4B, and model comparison experiments. **In progress.**

WP4: Voice Integration and Physical Deployment (May to Jul 2026). STT integration (Moonshine or system speech recognition), TTS integration (Piper or system TTS), physical FarmBot deployment over ROS2, and latency optimisation. **Planned.**

WP5: User Evaluation and Thesis Writing (Jun to Aug 2026). Ethics approval, evaluation sessions with elderly participants, thesis writing, poster, video, and presentation preparation. **Planned.**

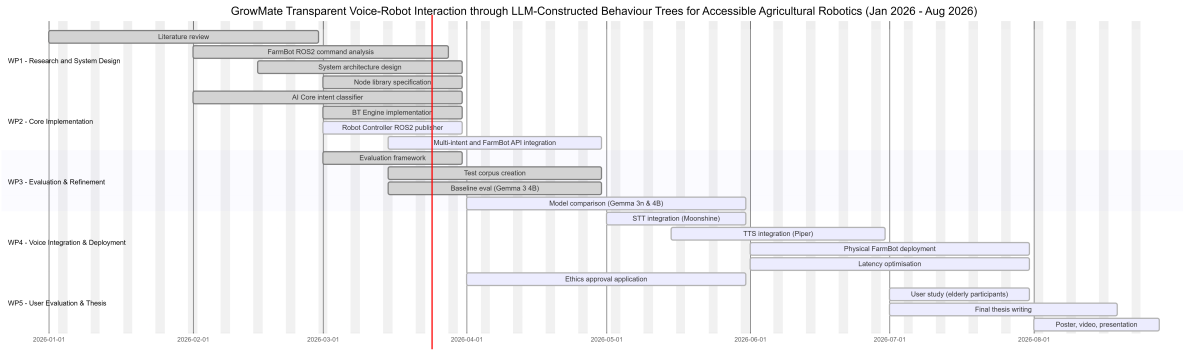

Figure 2: Project Gantt chart. Completed tasks in dark shading, planned tasks in light shading. Diamond markers indicate milestones.

4.2 Gantt Chart

Figure 2 shows the timeline for all work packages, tasks, and milestones.

4.3 Completed Work

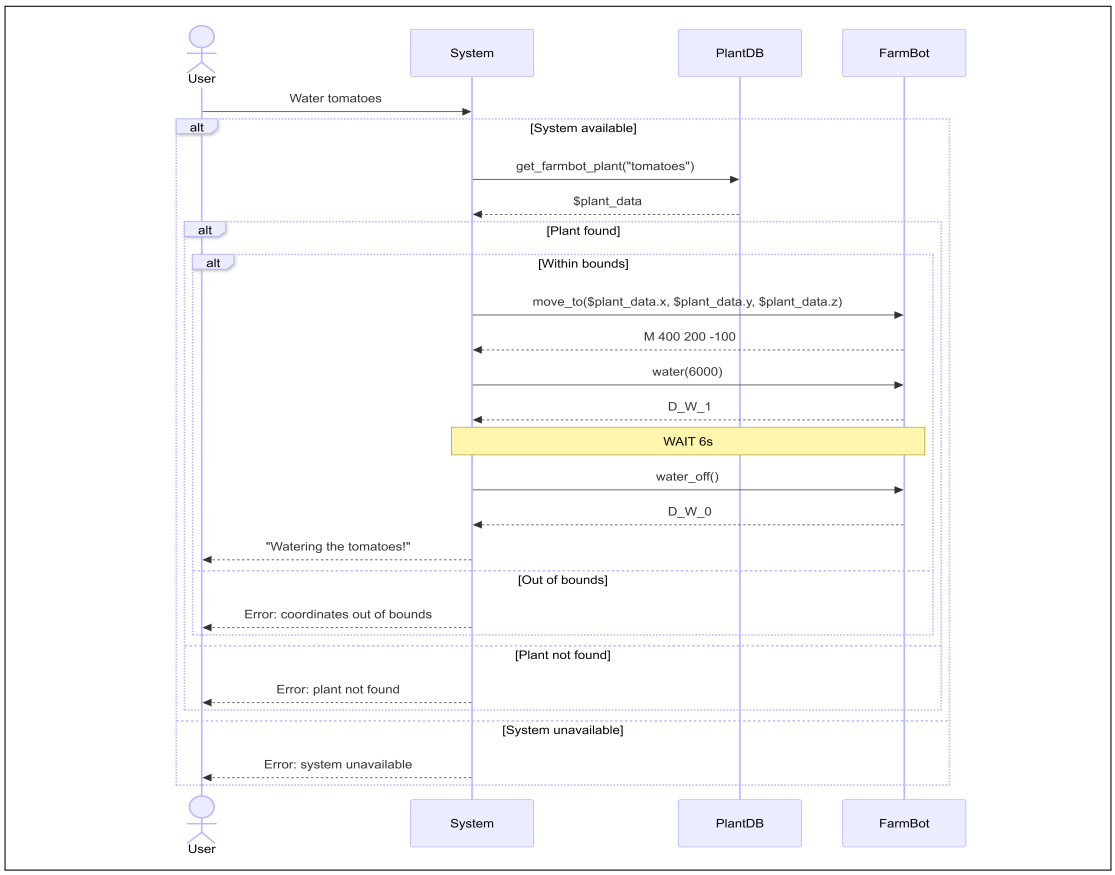

Figure 3: Sequence diagram for the "water the tomatoes" command, showing safety validation branches and FarmBot command generation.

A three-layer architecture has been designed and implemented following the high-level design document agreed with the supervisor. The User Interaction Layer handles speech input and

spoken feedback. The AI Processing Core classifies intent and builds behaviour trees. The Robot Control Layer manages communication with FarmBot via ROS2.

The BT engine, the node library, and the AI Core have been implemented as described in Section 3. Multi-intent support means that compound requests like "water the tomatoes and then go home" produce a single tree containing both sub-tasks. We had not expected this to work on the first attempt, but the flat JSON output format turned out to be simple enough that even a 4B model handles it reliably.

An evaluation framework has been implemented measuring DBSR, SNSR, USC, tree validity, tree dimensions, action granularity, and safety coverage across 29 test cases in 7 categories. A tree dump feature allows every generated tree to be inspected in full.

4.4 Preliminary Results

Table 1 summarises the functional results from running the 29 test utterances with Gemma 3 4B. Table 2 presents the non-functional metrics.

Table 1: Functional evaluation results by utterance category.

Category	**DBSR**	**Avg Latency**	**Avg Nodes**	**Safety**
Direct commands (10)	10/10	4,595 ms	5.3	Yes
Indirect commands (5)	5/5	8,289 ms	10.0	Yes
Sensor queries (4)	3/4	5,782 ms	6.2	Yes
General questions (3)	3/3	7,926 ms	5.7	N/A
Multi-step tasks (2)	2/2	10,236 ms	15.5	Yes
Safety-critical (1)	1/1	3,441 ms	5.0	Yes
Emergency (4)	4/4	<1 ms	3.0	Bypass
Overall (29)	**28/29**	**5,456 ms**	**6.8**	**76%**

Table 2: Non-functional evaluation metrics.

Metric	**Value**
Tree Validity Rate	29/29 (100%)
Single Node Success Rate	162/164 (98.8%)
Unsafe State Count	0

Twenty-eight of 29 utterances produced trees that achieved the intended goal. Every generated tree was structurally valid. Across 164 individual node executions, 162 succeeded. The unsafe state count was zero: no movement exceeded workspace bounds, no command targeted a nonexistent plant, and the one operation requiring confirmation was correctly gated. One utterance failed due to a classification ambiguity between `check_moisture` and `check_sensor` for a sensor query; the system checked the wrong sensor type, not a dangerous operation.

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